

$$\mu = 4e+05 \text{ km}^3/\text{sec}^2$$

$$\ddot{r} - r\dot{\lambda}^2 \cos^2 \lambda - r\dot{\lambda}^2 = -\mu/r^2 + F_r/m \dots \textcircled{a}$$

$$r\ddot{\lambda} \cos \lambda + 2\dot{r}\dot{\lambda} \cos \lambda - 2r\dot{\lambda}\dot{\lambda} \sin \lambda = F_\lambda/m \dots \textcircled{b}$$

$$r\ddot{\lambda} + 2\dot{r}\dot{\lambda} + r\dot{\lambda}^2 \sin \lambda \cos \lambda = F_\lambda/m \dots \textcircled{c}$$

$$r = 17400 \text{ km}, \lambda = 0, \vec{F} = 0 \Rightarrow F_r = F_\lambda = F_\lambda = 0$$

$$\text{Since } r = 17400 \text{ km} \Rightarrow \dot{r} = 0 \text{ km/s} \text{ \& } \ddot{r} = 0 \text{ km/s}^2$$

$$\text{Since } \lambda = 0, \dot{\lambda} = 0 \text{ rad/s} \text{ \& } \ddot{\lambda} = 0 \text{ rad/s}^2$$

$$\Rightarrow (17400 \text{ km})\ddot{\lambda}^2 = +\mu/(17400 \text{ km})^2 \text{ from } \textcircled{a}$$

$$(17400 \text{ km})\ddot{\lambda} = 0 \text{ from } \textcircled{b}$$

The both sides of $\textcircled{c}$ are zero.

$$\Rightarrow \ddot{\lambda}^2 = +\mu/(17400 \text{ km})^3 \text{ \& } \ddot{\lambda} = 0$$

$$\Rightarrow \dot{\lambda} = \sqrt{\frac{4e+05 \text{ km}^3/\text{sec}^2}{(17400 \text{ km})^3}} \approx 9.935 \text{ e-04 rad/s}$$

Thus, the constant orbital velocity $\vec{v} = r\dot{\lambda} = 7.352 \text{ km/s}$

The orbital period is $T = 2\pi/\dot{\lambda} \approx 6324.293 \text{ s} = 1.757 \text{ hours}$.

B. Let R be a radius of an arbitrary circular reference orbit

Since the orbit is circular, $\lambda^* = 0 = \dot{\lambda}^* = \ddot{\lambda}^*$ and $\dot{R} = 0 = \ddot{R}$

$$\text{From } \textcircled{a}, -R\dot{\lambda}^2 = -\mu/R^2 \Rightarrow \dot{\lambda} = \sqrt{\mu/R^3} \equiv \omega_0$$

$$\text{From } \textcircled{b}, R\ddot{\lambda} = 0 \Rightarrow \ddot{\lambda} = 0$$

The both sides of $\textcircled{c}$ are zero.

Hence $\dot{\lambda}(t) = \omega_0 t + C$ for some constant C .

Define $r = R + \Delta r$, $\lambda = 0 + \Delta \lambda$, and $\dot{\lambda} = \dot{\lambda}^* + \Delta \dot{\lambda}$ for equilibrium solutions $(R, \dot{\lambda}^*, 0)$.

From $\textcircled{a}$,

$$\begin{aligned} \text{Left side} &: (\ddot{R} + \ddot{\Delta r}) - (R + \Delta r)(\dot{\lambda}^* + \Delta \dot{\lambda})^2 \cos^2 \Delta \lambda - (R + \Delta r)\dot{\lambda}^2 \\ &= \ddot{\Delta r} - (R + \Delta r)(\dot{\lambda}^{*2} + 2\Delta \dot{\lambda}\dot{\lambda}^*) \\ &= \ddot{\Delta r} - R\omega_0^2 - \omega_0^2 \Delta r - 2R\omega_0 \Delta \dot{\lambda} \end{aligned}$$

$$\begin{aligned} \text{Right side} &: -\mu/(R + \Delta r)^2 + F_r/m \\ &= -(\mu/R^2)(1 + 2\frac{\Delta r}{R})^{-1} + F_r/m \\ &= -(\mu/R^2)(1 - 2\frac{\Delta r}{R}) \\ &= -\omega_0^2(R - 2\Delta r) + F_r/m \\ &= -\omega_0^2 R + 2\omega_0^2 \Delta r + F_r/m \end{aligned}$$

$$\Rightarrow \ddot{\Delta r} - 2R\omega_0 \Delta \dot{\lambda} - \omega_0^2 \Delta r - R\omega_0^2 = -\omega_0^2 R + 2\omega_0^2 \Delta r + F_r/m$$

$$\Rightarrow \ddot{\Delta r} - 2R\omega_0 \Delta \dot{\lambda} - 3\omega_0^2 \Delta r = F_r/m \dots \textcircled{d}$$

From $\textcircled{b}$

$$\begin{aligned} \text{Left side} &: (R + \Delta r)(\ddot{\lambda}^* + \ddot{\Delta \lambda}) \cos^2 \Delta \lambda + 2(R + \Delta r)(\dot{\lambda}^* + \Delta \dot{\lambda}) \cos \Delta \lambda - 2(R + \Delta r)\dot{\lambda}^* \sin \Delta \lambda \\ &= (R + \Delta r)\ddot{\Delta \lambda} + 2\Delta r(\omega_0 + \Delta \dot{\lambda}) \\ &= R\ddot{\Delta \lambda} + 2\omega_0 \Delta \dot{\lambda} \end{aligned}$$

$$\Rightarrow R\ddot{\Delta \lambda} + 2\omega_0 \Delta \dot{\lambda} = F_\lambda/m \dots \textcircled{e}$$

From (C),

$$\begin{aligned} \text{Left side : } (R+\Delta r) \ddot{\Delta \lambda} + 2(R+\Delta r) \dot{\Delta \lambda} \dot{\Delta \lambda}^0 + (R+\Delta r) (\dot{\Delta \lambda}^0 + \Delta \mu)^2 \sin^2 \Delta \lambda \cos^2 \Delta \lambda \\ = R \ddot{\Delta \lambda} + (R+\Delta r) (\omega_0^2 + 2\omega_0 \Delta \mu) \Delta \lambda \\ = R \ddot{\Delta \lambda} + R \omega_0^2 \Delta \lambda \end{aligned}$$

$$\Rightarrow R \ddot{\Delta \lambda} + R \omega_0^2 \Delta \lambda = F_r / m \dots \textcircled{4}$$

Define $x = \Delta r$, $y = R \Delta \mu$ and $z = R \Delta \lambda$.

$$\textcircled{4} : \ddot{x} - 2\omega_0 \dot{y} - \omega_0 x = F_r / m \dots \textcircled{4}'$$

$$\textcircled{5} : \ddot{y} - 2\omega_0 \dot{x} = F_y / m \dots \textcircled{5}'$$

$$\textcircled{6} : \ddot{z} + \omega_0 z = F_z / m \dots \textcircled{6}'$$

Those equations are called Hill-Clohesy-Wiltshire equations.

$$\#2. [x, y, z]^T = [-110, 70, 10]^T \text{ km.}$$

$$[\dot{x}, \dot{y}, \dot{z}]^T = [-10, 20, 1]^T \text{ km/s.}$$

Maneuvering time $T = 2h = 7200 \text{ s.}$

$$\text{Initial condition } [x_0, \dot{x}_0, y_0, \dot{y}_0, z_0, \dot{z}_0]^T = [-110, -0.01, 70, 0.02, 10, 0.001]^T$$

$$\text{Final condition } [x_f, \dot{x}_f, y_f, \dot{y}_f, z_f, \dot{z}_f]^T = [0, 0, 0, 0, 0, 0]^T$$

$R = 38500 \text{ km}$: the radius of the reference frame.

$$\omega = \sqrt{\frac{\mu}{R^3}} = \sqrt{\frac{4E+05 \text{ km}^3/\text{s}^2}{(38500 \text{ km})^3}} \approx 8.372E-05 \text{ rad/s.}$$

By the hint, $\omega T \approx 0.601$

$$\dot{y}_0^+ = \frac{[6x_0(\omega T - \sin \omega T) - y_0] \omega \sin(\omega T) - 2\omega x_0(4 - 3\cos \omega T)(1 - \cos \omega T)}{(4\sin \omega T - 3\omega T)\sin \omega T + 4(1 - \cos \omega T)^2}$$

$$\Rightarrow \text{Numerator : } 6x_0(\omega T - \sin \omega T) = -660(0.601 - \sin(0.601)) \approx -23.451$$

$$\omega \sin(\omega T) = 8.372E-05 \times \sin(0.601) \approx 4.734E-05$$

$$2\omega x_0(4 - 3\cos \omega T) = 2 \times 8.372E-05 \times (-110) \times (4 - 3\cos(0.601)) \approx -0.0281$$

$$1 - \cos \omega T = 1 - \cos(0.601) \approx 0.175$$

$$\Rightarrow \text{Numerator is } -4.424E-03 + 4.918E-03 = 4.935E-04$$

$$\Rightarrow \text{Denominator : } (4\sin \omega T - 3\omega T)\sin \omega T$$

$$= (4\sin(0.601) - 3 \times 0.601)\sin(0.601)$$

$$\approx 0.259$$

$$4(1 - \cos \omega T)^2 = 4(1 - \cos(0.601))^2$$

$$\approx 0.1228$$

$$\Rightarrow \text{Denominator is } 0.3818$$

$$\therefore \dot{y}_0^+ = +1.293E-03 \text{ km/s.}$$

$$\dot{x}_0^+ = - \frac{\omega x_0(4 - 3\cos(\omega T)) + 2(1 - \cos(\omega T)) \dot{y}_0}{\sin(\omega T)}$$

$$\Rightarrow \text{Numerator : } \omega x_0(4 - 3\cos(\omega T)) = 8.372E-05 \times (-110) \times (4 - 3\cos(0.601)) \approx -0.0141$$

$$2(1 - \cos(\omega T)) \dot{y}_0 = 2(1 - \cos(0.601)) (1.243 \times 10^{-3}) \\ \approx 4.531 \times 10^{-4}$$

⇒ Numerator is -0.0136

⇒ Denominator: $\sin \omega T = \sin(0.601) \approx 0.565$

∴ $\dot{x}_0^+ \approx 0.0242 \text{ km/s}$

$\dot{z}_0^+ = -z_0 \omega \cot(\omega T) = -10 \times 3.872 \times 10^{-5} \times \cot(0.601) \\ \approx -0.0306 \text{ km/s}$

⇒ $\Delta \vec{V}_0 = (\dot{x}_0^+ - \dot{x}_0) \vec{e}_r + (\dot{y}_0^+ - \dot{y}_0) \vec{e}_\mu + (\dot{z}_0^+ - \dot{z}_0) \vec{e}_\lambda \\ = (0.0242 + 0.01) \vec{e}_r + (1.243 \times 10^{-3} - 0.02) \vec{e}_\mu + (-0.0306 - 0.001) \vec{e}_\lambda \\ = [0.0342, -0.0187, -0.0316] \text{ km/s}$

⇒ $\Delta V_0 = \|\Delta \vec{V}_0\| = 0.0502 \text{ km/s}$

From Hint 3

$x(t) = \frac{\dot{x}_0^+}{\omega} \sin(\omega t) - (3x_0 + \frac{2\dot{y}_0^+}{\omega}) \cos(\omega t) + (4x_0 + \frac{2\dot{y}_0^+}{\omega})$

⇒ $\dot{x}(t) = \dot{x}_0^+ \cos(\omega t) + (3x_0 \omega + 2\dot{y}_0^+) \omega \sin(\omega t)$

⇒ $\dot{x}_F = \dot{x}(T) = \dot{x}_0^+ \cos(\omega T) + (3x_0 \omega + 2\dot{y}_0^+) \omega \sin(\omega T) \\ = 0.0242 \cos(0.601) + (3 \times 0.0242 \times 3.872 \times 10^{-5} + 2 \times 1.243 \times 10^{-3}) \omega \sin(0.601) \\ \approx 0.0200 + 2.592 \times 10^{-3} \times 4.734 \times 10^{-3} \\ = 0.0200 \text{ km/s}$

$y(t) = (6x_0 + \frac{4\dot{y}_0^+}{\omega}) \sin(\omega t) + \frac{2x_0}{\omega} \cos(\omega t) - (6\omega x_0 + 3\dot{y}_0^+)t + (y_0 - \frac{2x_0}{\omega})$

⇒ $\dot{y}(t) = (6\omega x_0 + 4\dot{y}_0^+) \cos(\omega t) - 2x_0 \sin(\omega t) - (6\omega x_0 + 3\dot{y}_0^+)$

⇒ $\dot{y}_F = \dot{y}(T) = (6\omega x_0 + 4\dot{y}_0^+) \cos(\omega T) - 2x_0 \sin(\omega T) - (6\omega x_0 + 3\dot{y}_0^+) \\ = (6 \times 3.872 \times 10^{-5} \times (-110) + 4 \times (1.243 \times 10^{-3})) \cos(0.601) \\ - 2 \times (0.0242) \sin(0.601) - (6 \times 3.872 \times 10^{-5} \times (-110) + 3 \times (1.243 \times 10^{-3})) \\ \approx -0.0168 - 0.0274 + 0.0217 \\ = -0.0225 \text{ km/s}$

$z(t) = \frac{\dot{z}_0^+}{\omega} \sin(\omega t) + z_0 \cos(\omega t)$

⇒ $\dot{z}(t) = \dot{z}_0^+ \cos(\omega t) - z_0 \omega \sin(\omega t)$

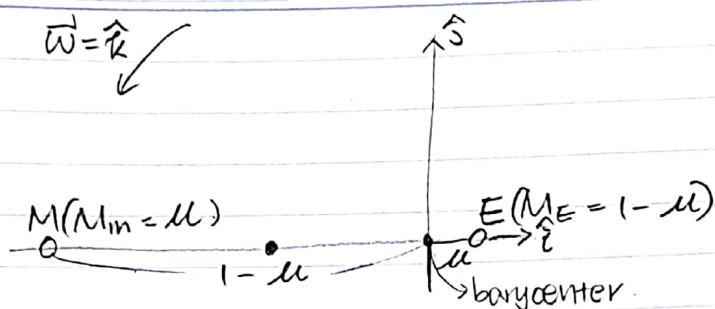
⇒ $\dot{z}_F = \dot{z}(T) = \dot{z}_0^+ \cos(\omega T) - z_0 \omega \sin(\omega T) \\ = -0.0306 \cos(0.601) - 10 \times 3.872 \times 10^{-5} \times \sin(0.601) \\ \approx -0.0252 - 2.189 \times 10^{-4} \\ = -0.0254 \text{ km/s}$

$\Delta \vec{V}_F = [-0.0200, 0.0225, 0.0254] \text{ km/s}$

⇒ $\Delta V_F = 0.0394 \text{ km/s}$

#3

$$\vec{w} = \hat{k}$$



$$\begin{cases} x_0 = \mu - 0.5 = -0.4877 \\ y_0 = 0 = z_0 \\ \dot{x}_0 = 0 = \dot{y}_0 = \dot{z}_0 \\ \mu = 0.0123 \end{cases}$$

A. If the spacecraft in equilibrium, each equations of motion for restricted 3 body system holds for the state of the spacecraft. ($\ddot{x} = \ddot{y} = \ddot{z} = 0$).

$$\ddot{x} - 2\dot{y} - x = -\frac{(1-\mu)(x-\mu)}{r_1^3} - \frac{\mu(x+1-\mu)}{r_2^3}$$

$$\Rightarrow 0 - 2 \cdot 0 + 0.4877 = -\frac{(1-0.0123)(-0.4877-0.0123)}{(0.4877+0.0123)^3} - \frac{0.0123(-0.4877+1-0.0123)}{(-0.4877+1-0.0123)^3}$$

$$\Rightarrow \text{Left side : } 0.4877$$

$$\text{Right side : } 3.9508 - 0.1449 = 3.8059$$

The left side and right side are totally different.

Hence the spacecraft is not in equilibrium.

The second and third equations hold because both sides are zero.

Thus there exist only $\hat{x}$ -axis component acceleration on the spacecraft w.r.t the barycenter frame.

The acceleration is,

$$\ddot{x} = 2\dot{y} + x - \frac{(1-\mu)(x-\mu)}{r_1^3} - \frac{\mu(x+1-\mu)}{r_2^3} \approx 3.8059 - 0.4877 = 3.3182$$

B. For keeping the spacecraft in equilibrium at this midway points make the first equations of motion hold.

The other equations of motion are already hold, then we consider only $\hat{x}$ -axis term.

$$\ddot{x} - 2\dot{y} - x = -\frac{(1-\mu)(x-\mu)}{r_1^3} - \frac{\mu(x+1-\mu)}{r_2^3} + \frac{F_x}{m}$$

$$\Rightarrow 0.4877 = 3.8059 + F_x/m$$

$$\Rightarrow F_x/m = 0.4877 - 3.8059 \approx -3.3182$$

Hence, the control acceleration is $-3.3182 \hat{x}$.

#4 $\Delta x \in \mathbb{R}^2$. $\Delta \dot{x} = A \Delta x$, $\Delta x(0) = \Delta x_0$. Let λ be eigenvalues of A .

Let $\lambda = \pm j\omega$ and $\xi, \bar{\xi}$ be eigenvectors corresponding to $j\omega, -j\omega$.

The solution of $\Delta \dot{x} = A \Delta x$ is,

$$\Delta x = e^{At} \Delta x_0 = a_1 \xi e^{j\omega t} + a_2 \bar{\xi} e^{-j\omega t} \text{ for some } a_1, a_2.$$

$$= a_1 \xi (\cos \omega t + j \sin \omega t) + a_2 \bar{\xi} (\cos \omega t - j \sin \omega t)$$

$$= (a_1 \xi + a_2 \bar{\xi}) \cos \omega t + j(a_1 \xi - a_2 \bar{\xi}) \sin \omega t$$

Since $\Delta x(0) = \Delta x_0$, $\Delta x_0 = a_1 \xi + a_2 \bar{\xi}$.

$$\text{Also, } \Delta \dot{x}_0 = j\omega(a_1 \xi - a_2 \bar{\xi})$$

$$\Delta x_0 = [\xi | \bar{\xi}] \begin{bmatrix} a_1 \\ a_2 \end{bmatrix}$$

$$\Rightarrow \Delta \dot{x}_0 = j\omega [\xi | -\bar{\xi}] \begin{bmatrix} a_1 \\ a_2 \end{bmatrix} = [\xi | \bar{\xi}] \begin{bmatrix} j\omega & 0 \\ 0 & -j\omega \end{bmatrix} \begin{bmatrix} a_1 \\ a_2 \end{bmatrix}.$$

Solve for (a_1, a_2) .

for Δx_0 composing a Halo Orbit, $\Delta x_0 \in \mathbb{R}^2$.

for $\Delta x_0 \in \text{Span}\{\text{Im}\xi, \text{Re}\xi\}$

It means, if $a_1 = a_2$, $\Delta x_0 = 2a_1 \text{Re}\xi$.

for example, if $a_1 = -a_2$, $\Delta x_0 = 2a_2 \text{Im}\xi$.